

Question				Marking details	Marks available																																								
					AO1	AO2	AO3	Total	Maths	Prac																																			
6	(a)			Is a [light] {string / thread} with a {bob / mass / weight} attached to one end	1			1		1																																			
	(b)			Acceleration is proportional to its {distance / displacement} (1) and directed towards its {equilibrium / fixed} point (1)	2			2																																					
	(c)			[System A] the angle is too big [for shm to occur] (1) Any × (1) from: - the shorter length {has a larger [percentage] uncertainty / is less accurate} - the shorter period {has a larger [percentage] uncertainty / is less accurate}			2	2		2																																			
	(d)	(i)		<table><tr><th>l / m</th><th>$\sqrt{l}$ / m^½</th><th>T_{20} / s</th><th>Mean T_{20} / s</th><th>Uncertainty T_{20} / s</th></tr><tr><td>0.250</td><td>0.50</td><td>19, 21, 22, 20</td><td>21</td><td>2</td></tr><tr><td>0.500</td><td>0.71</td><td>26, 29, 26, 29</td><td>28</td><td>2</td></tr><tr><td>0.750</td><td>0.87</td><td>35, 34, 33, 36</td><td>35</td><td>2</td></tr><tr><td>1.000</td><td>1.00</td><td>42, 39, 40, 41</td><td>41</td><td>2</td></tr><tr><td>1.250</td><td>1.12</td><td>45, 42, 44, 45</td><td>44</td><td>2</td></tr><tr><td>1.500</td><td>1.22</td><td>48, 50, 49, 52</td><td>50</td><td>2</td></tr></table> [1 for each column] Deduct a maximum of 1 mark overall for inconsistent sig figs in each column	l / m	$\sqrt{l}$ / m ^½	T_{20} / s	Mean T_{20} / s	Uncertainty T_{20} / s	0.250	0.50	19, 21, 22, 20	21	2	0.500	0.71	26, 29, 26, 29	28	2	0.750	0.87	35, 34, 33, 36	35	2	1.000	1.00	42, 39, 40, 41	41	2	1.250	1.12	45, 42, 44, 45	44	2	1.500	1.22	48, 50, 49, 52	50	2		3		3	3	3
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		(ii)		See example graphs at end of Q6 Award 3 marks for all points and error bars correct Award 2 marks for 5 points and error bars correct Award 1 mark for 4 points and error bars correct Award 0 marks for 3 points or less and error bars correct Correct lines drawn for the maximum gradient and the minimum gradient (1) ecf		4		4	4	4
		(iii)		l can be measured to ± 0.001 [m] or resolution is 1 mm (1) [Uncertainty in $\sqrt{l}$] is too small to plot (1)			2	2		2
		(iv)		Both gradients correct (1) Mean gradient correct (1) ecf Absolute uncertainty in gradient correct (1) ecf implied by % uncertainty Correct g (1) ecf Final % correct (1) ecf No sig fig or unit penalties $m_{\max} \left[= \frac{58.2}{1.40 - 0.06} \right] = 43.4; m_{\min} \left[= \frac{54.3 - 4.7}{1.4} \right] = 35.4$ $\therefore \text{Gradient, } m = 39.4 \pm \{4.0 / 10\}\%$ Either For 20 oscillations, $T = 20 \times 2\pi \sqrt{\frac{l}{g}}$ $\therefore \sqrt{g} = \frac{40\pi}{m} \text{ [or by impl.] [= } 3.19 \pm 10\%]$ $\therefore g = 10.2 \text{ m s}^{-2} \pm 20\%$		5		5	5	5

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				Or For one oscillation: $m = 1.97 \pm 10\%$ (1) $\left[T = 2\pi\sqrt{\frac{l}{g}} \right], \therefore \sqrt{g} = \frac{2\pi}{m}$ [or by impl.] [= $3.19 \pm 10\%$] (1) $\therefore g = 10.2 (\pm 20\%) \text{ m s}^{-2}$ Alternative: calculate max value of g from min gradient; min value of g from max gradient and hence $g \pm$ uncertainty. Gradients of both: 43.284 and 35.357 (1) Mean $T_{20} = \frac{40\pi}{\sqrt{g}} \sqrt{l}$ so gradient = $\frac{40\pi}{\sqrt{g}}$ $g = \left(\frac{40\pi}{43.3}\right)^2 = 8.42 \text{ m s}^{-2}$ and $g = \left(\frac{40\pi}{35.3}\right)^2 = 12.67 \text{ m s}^{-2}$ (1) So $\text{unc}(g) = \frac{12.67-8.42}{2} = 2.1 \text{ m s}^{-2}$ (1) $g = 10.5 \text{ m s}^{-2}$ [1] $\therefore \% \text{ unc}(g) = \left[\frac{2.1}{10.5} \times 100\% \right] = 20\%$ (1)						

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	(e)			Slope $\propto \frac{1}{\sqrt{g}}$ or equivalent e.g. gradient = $\frac{2\pi}{\sqrt{g}}$ or gradient = $\frac{40\pi}{\sqrt{g}}$ (1) Ratio of slopes = $\frac{1}{\sqrt{0.2}}$ or $\sqrt{5}$ or new gradient = 89 or two correct expressions for gradients e.g. $\frac{2\pi}{\sqrt{g}}$ and $\frac{2\pi}{\sqrt{0.2g}}$ or $\frac{40\pi}{\sqrt{g}}$ and $\frac{40\pi}{\sqrt{0.2g}}$ (1) Ratio of gradients = 2.24ish and sensible comment (1)			3	3	3	
				Question 6 total	3	12	7	22	15	17

